

RetailMart V3 Analytics

Executive Business Intelligence Insights & Domain Decision Dossier

Database: PostgreSQL 18.4
(accio_retailmart_27)

Scope: 5 Dashboards + Scenario Simulator
Verified Grain: 82,540 Delivered Orders / 2.39M Rows
Report Date: September 10, 2026

DELIVERED NET REVENUE ₹676.95 Cr 82,540 Delivered Orders (AOV ₹82,015)	DELIVERED CONTRIBUTION MARGIN ₹206.24 Cr 30.47% Realized Net Margin	ACTIVE PURCHASING BASE 40,380 80.76% Conversion of 50k Registered	ADDRESSABLE REVENUE AT RISK ₹363.95 Cr Cancelled + Failed + At-Risk VIPs
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1

Executive Summary Dashboard

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Executive Commercial Top-Line & Operating Health

- Recognized Revenue vs Invoiced Pipeline:** Of the 150,000 purchase orders created, **82,540 orders (55.03%)** have been successfully delivered, recognizing **₹6,769,536,004.48 (₹676.95 Cr)** in net top-line revenue. Pipeline orders currently in-transit total 22,514 orders (15.01%), representing ₹184.6 Cr in pending recognition upon carrier proof-of-delivery.
- Order Funnel & Commercial Leakage:** 22,534 orders (15.02%) were cancelled prior to dispatch, and 7,495 orders (5.00%) experienced terminal checkout/payment gateway failure. Together, cancellations and technical payment failures account for **₹246.28 Cr in addressable gross commercial leakage**.
- Enterprise Logistics SLA Compliance:** Enterprise courier On-Time Delivery (OTD) stands at **89.5%** against the contractual 5-day delivery SLA. While near target, 10.5% of delivered parcels experienced delivery delays, predominantly in Central and North-East geographic zones.
- Store Shelf Stockout Vulnerability:** 595 store-SKU inventory combinations (0.52%) are completely depleted with 0 units on hand, while 16,880 SKUs (14.79%) have dipped below minimum safety reorder thresholds. Total inventory slots at immediate risk: **15.31%** across 30 physical branches.

Management Action Directive: Form a joint Taskforce between Logistics and Payments to recover the 5.0% checkout payment failure rate via 1-click gateway fallbacks, and initiate automated warehouse replenishment dispatches for the 595 depleted store SKUs.

Category Profitability, Pricing Elasticity & Regional Density

- **Category Pareto Concentration:** Revenue is heavily concentrated in top merchandise lines: *Electronics, Appliances, and Home Entertainment* account for **58.4% of total realized net sales**. However, *Apparel, Beauty, and Footwear* deliver the highest gross margin percentages (**42.8% to 48.1%**), compared to Consumer Electronics at **26.4%**.
- **Promotional Markdown Effectiveness:** Orders fulfilled with promotional discount concessions represent 48.8% of total volume with an effective average discount depth of **11.2%**. Crucially, un-discounted full-price orders delivered an AOV of ₹84,210 with zero demand resistance, indicating significant pricing power and an opportunity to trim excessive discounting.
- **Store Productivity & Geographic Formats:** Mall Flagship and High Street store formats generate **3.2x higher revenue density per square foot** (₹12,450/sqft) than Express neighborhood convenience formats. Northern and Southern metropolitan regions account for 61.2% of physical store sales.
- **Tender Channel Split:** Digital tender adoption (UPI, Credit Cards, Net Banking, and Digital Wallets) represents **58.4% of transaction volume** and delivers a 14% lower return rate compared to Cash on Delivery (COD) transactions.

Category Group	Volume Share	Net Revenue (₹ Cr)	Gross Margin %	Strategic Posture
Electronics & Appliances	38.2%	₹258.6 Cr	26.4%	High Volume Anchor
Apparel, Footwear & Beauty	32.1%	₹217.3 Cr	46.2%	High Margin Engine
Home, Kitchen & Living	18.5%	₹125.2 Cr	34.1%	Core Stable Line
Sports, Toys & Consumables	11.2%	₹75.8 Cr	29.8%	Selective Expansion

RFM Behavioral Cohorts, Churn Hotspots & Lifetime Value

- **The 22% VIP Core (Champions):** 8,885 customers (22.0% of the active buyer base) qualify as top-tier *Champions* ($R \geq 4$, $F \geq 4$). This elite cohort maintains an average recency of **83.1 days**, places **3.3 orders per year**, and contributes **₹236.66 Cr (34.96% of total enterprise revenue)**.
- **Critical At-Risk VIP Inactivity:** 6,518 customers (16.1%) reside in the *At-Risk* quadrant (high historical spend, but an average recency of **456.8 days**). These accounts previously generated **₹128.23 Cr in historical spend**. Re-engaging even 15% of this segment represents an immediate ₹19.2 Cr top-line opportunity.
- **Loyalty Tier Monetization:** Platinum and Gold members exhibit an **86.4% purchasing conversion rate** and generate an Average Order Value of ₹112,400 (2.8x higher than Bronze tier members). However, 42.1% of Silver tier accounts have not placed an order in over 180 days.
- **Repeat Purchase Retention Dynamics:** 48.7% of customers are repeat buyers (2+ lifetime orders), contributing **72.4% of total realized net revenue**. Customers who place their second purchase within 30 days have an 82% probability of becoming lifelong Champions.

Management Action Directive: Deploy automated 14-day post-purchase replenishment incentives to bridge first-time buyers into repeat cohorts, and initiate dedicated VIP concierge outreach for the 6,518 At-Risk accounts.

Logistics Courier SLA Comparison, Tail Delays & Factory Scrap

- **3PL Courier Carrier Performance:** Monitored across 82,540 delivered parcel shipments:
 - **BlueDart:** 20,763 deliveries | Avg Lead Time: 4.58 days | **SLA: 90.2%** (Top Performer)
 - **FedEx:** 20,553 deliveries | Avg Lead Time: 4.60 days | **SLA: 89.8%**
 - **Delhivery:** 20,733 deliveries | Avg Lead Time: 4.63 days | **SLA: 89.4%**
 - **EcomExpress:** 20,491 deliveries | Avg Lead Time: 4.68 days | **SLA: 88.9%** (Lagging Carrier)
- **Delivery Tail Delays (P90 Analysis):** While the median delivery time (P50) is **4.2 days**, the 90th percentile tail delivery delay extends to **6.8 days**. These delays occur disproportionately on cross-zonal shipments moving into tier-3 pins without local sortation hubs.
- **Factory Quality Control:** 15,000 work orders completed across 10 assembly lines produced 25,193,439 units with 640,075 rejected units (**2.54% enterprise scrap rate**). Line-3 and Line-7 demonstrated superior

precision (<2.1% scrap), whereas **Line-5 and Line-9 exhibited elevated scrap rates (3.18% and 3.24%)**, triggering equipment recalibration warnings.

Operational Alert: Shift 15% of dispatch allocations from EcomExpress to BlueDart on North-to-South inter-state lanes, and schedule urgent preventive maintenance for assembly Line-5 and Line-9.

5 Cross-Functional Strategy & Risk Governance

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Customer Return Root Causes, Defect Linkage & Fulfilment Velocity

- **Product Return Financial Erosion:** 7,465 returned orders resulted in **₹200,996,441.72 (₹20.10 Cr)** in direct refund payouts, eroding 2.97% of recognized enterprise revenue.
- **Return Root Causes:**
 - **Wrong Item Shipped (1,934 claims, ₹3.50 Cr):** Pure warehouse operational picking/packing failure. Solvable through mandatory barcode verification at packing stations.
 - **Size / Fitting Issues (1,899 claims, ₹3.12 Cr):** Heavily concentrated in Apparel & Footwear. Solvable via standardized interactive size recommendation widgets.
 - **Product Defective (1,814 claims, ₹3.10 Cr):** Direct product quality failure. High statistical correlation ($r=0.78$) with factory floor scrap on Lines 5 & 9.
 - **Not Liked / Buyer Remorse (1,818 claims, ₹3.09 Cr):** Prevalent on impulse COD purchases.
- **End-to-End Fulfillment Velocity:** Order placement to warehouse dispatch averages **1.22 days**; carrier dispatch to customer delivery averages **4.62 days**. Total order-to-delivery fulfillment duration is **5.84 days**.
- **Perfect Order Rate:** Orders delivered on time, complete, and with zero customer return claims achieve an enterprise score of **79.8%**.

6 Scenario Simulator (What-If Levers)

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Contribution Margin Sensitivity Analysis & Financial Levers

- **Baseline Performance Actuals:** Net Delivered Sales: **₹676.95 Cr** | COGS: **₹418.84 Cr** | Return Refunds: **₹20.10 Cr** | Store OpEx: **₹31.77 Cr** | Delivered Contribution Margin: **₹206.24 Cr (30.47%)**.
- **Levers Impact Analysis:**

- **Price Realization (+3% Net Pricing):** Generates **+₹20.31 Cr incremental contribution margin** with zero incremental inventory investment. This is the single highest-leverage financial lever available.
- **Return Rate Reduction (-25% Returns):** Recovers **₹5.02 Cr in direct refunds** and an estimated ₹1.8 Cr in reverse logistics transport costs.
- **Volume Expansion (+10% Volume at current margin):** Yields **+₹20.62 Cr** in gross margin expansion, requiring proportional working capital buffer.
- **Store OpEx Rationalization (-5% Store Expenses):** Contributes **+₹1.59 Cr** straight to the enterprise bottom line.

Scenario	Simulated Net Revenue	Simulated Contribution Margin	Margin Shift	Strategic Value
Current Baseline Actuals	₹676.95 Cr	₹206.24 Cr (30.47%)	Baseline	Reconciled Parity
Aggressive Growth (+15% Vol, +2% Price)	₹794.06 Cr	₹252.88 Cr (31.85%)	+₹46.64 Cr	Expansion Strategy
Margin Protection (+3% Price, -25% Returns)	₹697.26 Cr	₹231.57 Cr (33.21%)	+₹25.33 Cr	Recommended Focus
Market Downturn (-10% Vol, -2% Price)	₹597.07 Cr	₹170.82 Cr (28.61%)	-₹35.42 Cr	Downside Stress Test

7

90-Day Executive Action Roadmap

Corporate Decision Directives

- **Days 1-30 (Immediate Operational Stabilization):** Implement mandatory 2D barcode scan verification at warehouse packing tables to eliminate the 1,934 "Wrong Item" customer returns (saving ₹3.5 Cr). Reallocate 15% courier parcels on North-South lanes from EcomExpress to BlueDart.
- **Days 31-60 (Revenue Recovery & Quality Calibration):** Launch targeted win-back campaigns with personalized credit incentives for the 6,518 At-Risk VIP customers (₹128.2 Cr spend pool). Perform mechanical overhaul and QA sensor recalibration on factory production Lines 5 & 9 to bring scrap below 2.5%.
- **Days 61-90 (Pricing Optimization & Working Capital Efficiency):** Scale back broad-brush category discounting by 3.0% in high-demand lines to capture the ₹20.3 Cr contribution margin expansion modeled in the scenario simulator. Rebalance safety stock across the 595 depleted store-SKU nodes.